

Closing ROBUSTNESS-MU2: the STEP-5B beyond-layer margins re-computed off-anchor stay above unity across the neighbourhood

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

This note closes ROBUSTNESS-MU2 by settling its last residual – the exact off-anchor re-margining of the STEP-5B beyond-layer closure. Combined with the A=0-uniqueness robustness (note robustness-mu2-offanchor), the Reading-H selection is established robust on the $\mu^2 \in [\times 0.5, \times 2]$ neighbourhood. The exact layer margin $m(\mu^2)$ is bounded (not recomputed); this is a <few-% effect that cannot break the $\times 2.55$ endpoint floor (section 4).

2. The off-anchor re-margin

The AddE de-thinned closure margin is

$$\text{ratio}(\mu^2, I) = \frac{K_{\text{budget}}}{K(n_{\text{pack}})} = \frac{4(1 - a_0) m / [(\lambda' I)^2 J_{\text{eff}}]}{8 + 4\sqrt{14} \sqrt{n_{\text{pack}}}},$$

with $\lambda'(\mu^2) = 3u + 30vM_R$, $\hat{r}(\mu^2) = r_R + 2\lambda' I$, $a_0 = 2\lambda' I / \hat{r}$, $\theta_{\min} = \sqrt{\hat{r}} / (2q_0^2 \sqrt{C})$, $n_{\text{pack}} = 16 / \theta_{\min}^2$, and the minimum-transfer envelope $J_{\text{eff}} = J(2q_0 \sin(\theta_{\min}/2))$ recomputed at the off-anchor $\hat{r}$. Everything except J_{eff} is closed-form; the single envelope integral per (μ^2, I) is the only cost.

μ^2 (multiple)	ratio @ 4×10^{-4}	ratio @ 10^{-3}	ratio @ 2×10^{-3}
0.0025 ($\times 0.5$)	$\times 58.7$	$\times 9.7$	$\times 2.55$
0.00354 ($\times 0.71$)	$\times 59.0$	$\times 9.7$	$\times 2.56$
0.005 (anchor)	$\times 59.4$	$\times 9.8$	$\times 2.58$
0.00707 ($\times 1.41$)	$\times 59.9$	$\times 9.9$	$\times 2.60$
0.01 ($\times 2$)	$\times 60.8$	$\times 10.0$	$\times 2.64$

The worst ratio over the whole neighbourhood and all intensities is $\times 2.55$ (the $\times 0.5$, $I = 2 \times 10^{-3}$ corner) – the closure holds throughout with room. The margins are monotone increasing in μ^2 , so the anchor sits near the thinnest point.

3. The layer-margin floor is preserved

The ratio uses the anchor layer margin $m = 4.32 \times 10^{-3}$. Its own μ^2 -robustness is the Prop-A P_B floor: $M_R/M_c > 4.11$ throughout the neighbourhood (robustness-mu2-offanchor, 9/9), so the floor condition that makes $m > 0$ is preserved, and $m(\mu^2)$ stays positive and $O(\text{anchor})$ as the constants drift $< 2\%$.

4. Why the un-recomputed $m(\mu^2)$ cannot break closure

The endpoint floor is $\times 2.55$. The ratio scales linearly in $m(\mu^2)$; the constants entering m (via $P_B(M_R) - \text{dip}_B$) drift $< 2\%$ across the $\times 4$ band, so $m(\mu^2) = m_{\text{anchor}}(1 + O(2\%))$. Even a conservative 10% drop in m leaves the endpoint at $\times 2.3 > 1$. A break would require m to fall by $> 60\%$, impossible while $M_R/M_c > 4$ holds. The closure is therefore robust to the exact $m(\mu^2)$ variation; recomputing m exactly would only sharpen the already-comfortable margin.

5. Numerical verification

Script `codes/vacuum/robustness_mu2_step5b_remargin.py` (v1.0.0; 5/5 self-test asserts PASS; artefact `claims/B1-RH-ENUM/runs/260606-robustness-mu2-step5b/result.json`).

Quantitative sanity check (mandatory): the anchor re-margin reproduces the AddE floors ($\times 59.4$ at $I = 4 \times 10^{-4}$, $\times 2.6$ at $I = 2 \times 10^{-3}$; the $\times 9.8$ vs AddE $\times 8.8$ at $I = 10^{-3}$ is a J_{eff} -quadrature-resolution difference of $\sim 10\%$, which does not affect closure), confirming the re-margin shares the certified machinery; the endpoint flatness ($\times 2.55 - \times 2.64$) is consistent with the $< 2\%$ constant drift.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3.5)

- (α) “*You closed the gate using the ANCHOR m , not $m(\mu^2)$ – is the closure real?*” ADDRESSED in section 4: the $\times 2.55$ floor tolerates any $< 60\%$ drop in m ; the constants drift $< 2\%$ and $M_R/M_c > 4$ keeps $m > 0$, so $m(\mu^2)$ cannot break it. The no-overclaim states m is bounded, not recomputed.
- (β) “ *J_{eff} at $nk = 500$ gives $\times 9.8$ vs AddE $\times 8.8$ – is the quadrature trustworthy?*” VALID-with-mitigation: a $\sim 10\%$ envelope-resolution difference at one intensity; it does not change any closure verdict (all ratios > 1 with the thinnest at $\times 2.55$); the anchor endpoint $\times 2.6$ matches AddE.
- (γ) “*Why $[\times 0.5, \times 2]$ and not wider?*” The STEP-5B re-margin was computed on $[\times 0.5, \times 2]$ (a $\times 4$ band); the A=0-uniqueness component is robust wider ($[\times 0.2, \times 10]$). The gate closes for the intersection, $[\times 0.5, \times 2]$, stated explicitly; wider STEP-5B verification is a trivial extension if ever needed.
- (δ) “*Does closing the gate overstate B1?*” DISMISSED: the gate closes for a stated neighbourhood, not unconditionally in μ^2 ; B1’s scope records the $[\times 0.5, \times 2]$ robustness explicitly. No claim beyond it.

No objection upheld; ROBUSTNESS-MU2 is closed for the stated neighbourhood.

7. Result footer

Result ID:	B1-RH-ENUM / ROBUSTNESS-MU2 closure
Precise statement:	STEP-5B closure margin ratio = $K_{\text{budget}}/K(n_{\text{pack}})$ recomputed off-anchor stays > 1 across μ^2 in $[\times 0.5, \times 2]$ and all 3 intensities; worst $\times 2.55$ (endpoint), production-intensity $\sim \times 59$; margins monotone increasing in μ^2 (anchor near thinnest).

With $A=0$ uniqueness robust on $[x0.2, x10]$ and the Prop-A floor preserved ($M_R/M_c > 4.1$), ROBUSTNESS-MU2 CLOSED for $[x0.5, x2]$.

Scope: μ^2 in $[0.0025, 0.01]$ ($x0.5..x2$). Layer margin m bounded positive (P_B floor), not recomputed exactly (<few% effect, cannot break $x2.55$).

Dependencies: robustness-mu2-offanchor ($A=0$ + constants); AddE de-thinned margin; A1-KERNEL-CONV.

Evidence grade: ANALYTIC (closed-form ratio) + EXECUTED (5/5)

Reproduction command: `python codes/vacuum/robustness_mu2_step5b_remargin.py`

Expected output: claims 5/5 PASS; off-anchor ratios $x2.55..x60.8$; anchor reproduces AddE $x59.4/x2.6$; result.json under claims/B1-RH-ENUM/runs/260606-robustness-mu2-step5b/

Falsification gate: A μ^2 in $[x0.5, x2]$ with margin ratio ≤ 1 ; a layer margin $m(\mu^2)$ dropping $> 60\%$ (would need $M_R/M_c < \sim 1.6$, excluded by the sweep).

Tier before / after: ROBUSTNESS-MU2 OPEN $\rightarrow$ CLOSED (μ^2 in $[x0.5, x2]$). Removed from B1 open_gates. No tier-number change; B1 T6 scope gains neighbourhood robustness.

No-overclaim statement: Closure is for the STATED neighbourhood $[x0.5, x2]$, not unconditional in μ^2 . The exact layer margin $m(\mu^2)$ is bounded via the preserved P_B floor, not recomputed; this cannot break the $x2.55$ endpoint margin. No tier-number change.

Next required action: (optional) recompute $m(\mu^2)$ exactly to sharpen; remaining B1 open gates: ESTIMATOR-UPGRADE, G3PB-III; and SC-SCOPE all-orders.